

From Riemann Sums to the Fundamental Theorem

Answer Key

AP Calculus AB/BC

Quick Answers

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|---------------------------------------|--|
| 1. $\frac{15}{4}$ | 7. (a) trapezoidal estimate = 66, — |
| 2. 90 L | 8. (a) net change = 16, (b) amount at $t = 8 = 56$, (c) time of the maximum = 5 |
| 3. 26 | 9. (a) critical x-values = 2, 4, — |
| 4. 63 m | 10. (a) $A(9) = 108$ L, (b) $A'(4) = 12$, — |
| 5. (a) left sum = $\frac{77}{60}$, — | |
| 6. 204 | |
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What is verified

13 of 17 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 5, 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-6 – problems 3, 6, 9, 10

LIM-5 – problems 1, 2, 4, 5, 7, 8

Difficulty 1–5 of 5 across 17 tagged checks.

1. Right Riemann sum, $n = 4$, for $\int_0^2 x^2 dx$.

Solution. $\Delta x = \frac{2-0}{4} = \frac{1}{2}$ and the right endpoints are $\frac{1}{2}, 1, \frac{3}{2}, 2$, so the sum is

$$\frac{1}{2} \left(\frac{1}{4} + 1 + \frac{9}{4} + 4 \right) = \frac{1}{2} \cdot \frac{15}{2}.$$

$$\boxed{\frac{15}{4}}$$

That is 3.75, an overestimate of the exact $\frac{8}{3}$ because x^2 is increasing here.

2. $r(t) = 4t + 3$ L/min. Total volume in the first 6 minutes.

Solution. The total is the accumulated rate: $\int_0^6 (4t + 3) dt = [2t^2 + 3t]_0^6 = 72 + 18$.

$$\boxed{90 \text{ L}}$$

3. $G(x) = \int_2^x (t^3 - 1) dt$. Find $G'(3)$.

Solution. By the Fundamental Theorem $G'(x) = x^3 - 1$, so $G'(3) = 27 - 1$.

$$\boxed{G'(3) = 26}$$

No antiderivative was needed — that is the theorem doing the work.

4. $v(t) = 3t^2$ m/s. Displacement from $t = 1$ to $t = 4$.

Solution. Displacement is the integral of velocity: $\int_1^4 3t^2 dt = [t^3]_1^4 = 64 - 1$.

$$\boxed{63 \text{ m}}$$

5. $\int_1^3 \frac{1}{x} dx$. (a) Left sum, $n = 4$. (b) Over or under?

Solution (a). $\Delta x = \frac{1}{2}$ and the left endpoints are $1, \frac{3}{2}, 2, \frac{5}{2}$:

$$\frac{1}{2} \left(1 + \frac{2}{3} + \frac{1}{2} + \frac{2}{5} \right) = \frac{1}{2} \cdot \frac{77}{30}.$$

$$\boxed{\frac{77}{60}}$$

Solution (b) — instructor-judged. $\frac{1}{x}$ is decreasing on $[1, 3]$, so on each subinterval the left endpoint is the highest point: every rectangle pokes out above the curve and the sum overestimates. (The exact value is $\ln 3 \approx 1.0986$, well below $\frac{77}{60} \approx 1.2833$.) Full credit needs the link from “decreasing” to “rectangles too tall”, not just the word overestimate.

6. $H(x) = \int_1^{x^3} (2t + 1) dt$. Find $H'(2)$.

Solution. The upper limit is x^3 , so the chain rule supplies $3x^2$:

$$H'(x) = (2x^3 + 1) \cdot 3x^2.$$

At $x = 2$: $(16 + 1) \cdot 12 = 17 \cdot 12$.

$$\boxed{H'(2) = 204}$$

7. Speed table at $t = 0, 2, 4, 6, 8$: $v = 0, 5, 9, 12, 14$. (a) Trapezoidal estimate of $\int_0^8 v dt$. (b) Units and meaning.

Solution (a). Each subinterval is 2 wide, and each trapezoid has area $\frac{1}{2}(\text{left} + \text{right}) \cdot 2$:

$$2 \left(\frac{0+5}{2} + \frac{5+9}{2} + \frac{9+12}{2} + \frac{12+14}{2} \right) = 2(2.5 + 7 + 10.5 + 13).$$

66

Solution (b) — instructor-judged. The rate is in metres per second and the widths are in seconds, so the product is in metres; the estimate is the distance the cart travelled during those eight seconds. Full credit needs both the unit and the physical reading — a unit with no interpretation, or an interpretation with no unit, is half an answer.

8. Reservoir, $r(t) = 10 - 2t$ thousand litres per hour, 40 thousand litres at $t = 0$. (a) Net change. (b) Amount at $t = 8$. (c) Time of the maximum.

Solution (a). $\int_0^8 (10 - 2t) dt = [10t - t^2]_0^8 = 80 - 64$.

16

Thousand litres.

Solution (b). Starting amount plus net change: $40 + 16$.

56

Solution (c). The amount grows while $r > 0$ and shrinks after, so the peak is where $10 - 2t = 0$.

$t = 5$

There the reservoir holds $40 + 25 = 65$ thousand litres, more than at $t = 8$.

9. $F(x) = \int_0^x f$, $f(t) = t^2 - 6t + 8$, on $[0, 6]$. (a) Local extrema of F . (b) Which is the maximum?

Solution (a). $F' = f$, and $f(t) = (t - 2)(t - 4)$ vanishes at

$x = 2$ or $x = 4$

Solution (b) — instructor-judged. f is positive on $[0, 2)$, negative on $(2, 4)$ and positive on $(4, 6]$. Since $F' = f$, that makes F rise, fall, and rise again: $x = 2$ is a local maximum and $x = 4$ a local minimum. Full credit requires the sign change of f — F itself is never computed here, which is the point of the problem.

10. $r(t) = 6\sqrt{t}$ L/min on $[0, 9]$, $A(x) = \int_0^x 6\sqrt{t} dt$. (a) $A(9)$. (b) $A'(4)$. (c) What each one answers.

Solution (a). $\int 6\sqrt{t} dt = 4t^{3/2}$, so $A(9) = 4 \cdot 27 = 108$.

$A(9) = 108 \text{ L}$

Solution (b). By the Fundamental Theorem $A'(x) = 6\sqrt{x}$, so $A'(4) = 6 \cdot 2$.

$A'(4) = 12$

Solution (c) — instructor-judged. $A(9)$ is a *total*: the litres that arrived over the whole nine minutes, found by evaluating an antiderivative at the two limits. $A'(4)$ is a *rate*: the litres per minute arriving at the single instant $t = 4$, which the Fundamental Theorem hands back as the integrand itself. Full credit needs the total-versus-rate distinction with the right unit on each.